



ENTERPRISE RESOURCE PLANNING

An AI-powered Enterprise Resource Planning platform for centralized business management, analytics and intelligent decision-making.

Built for the Growth Marketing Community

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01. PROJECT AT A GLANCE

Enterprise Resource Planning is an **AI-powered business management system** designed to help organizations manage sales, inventory, finance, customers, employees and business reports through one centralized platform. It integrates with Zoho Books as the primary source of business data and uses Google Gemini AI to generate business insights and forecasts.

By centralizing operational information, the platform supports faster data-driven decision-making and reduces manual business analysis.

Core Modules

Module	Scope
Sales & Finance	Management of orders, invoices, payments, revenue, and financial forecasting.
Inventory & Suppliers	Tracking products, stock levels, warehouses, and supplier performance.
Customer & Employee	CRM capabilities and HR management for departments, attendance, and payroll.
AI Insights	Gemini-powered business health scores, demand forecasting, and recommendations.
Integrations	Direct connectivity with Zoho CRM, Books, and Inventory through OAuth.

Module	Scope
Global Search	Instant retrieval of invoices, customers, employees, and inventory records.

02. PROBLEM, USERS & PRODUCT VALUE

The Problem

Many businesses manage sales, inventory, finance, and reports across different systems. This creates fragmented data, repetitive manual work, delayed reporting and difficulty monitoring business performance from one place.

Target Users

- **Administrators:** Manage overall business systems, users, roles, and organization-level settings.
- **Finance Teams:** Review financial data, expenses, payments, invoices, and tax information.
- **Inventory Teams:** Track products, stock levels, warehouses, and low-stock alerts.

Product Value Proposition

The platform delivers value through a centralized AI-enhanced business strategy:

- **Centralized Business Management:** Consolidate sales, inventory, finance, and HR workflows into one unified platform to eliminate data fragmentation.
- **Real-time Business Dashboards:** Access live business KPIs, revenue trends, and operational metrics instantly for a comprehensive view of business health.
- **AI-Powered Insights:** Leverage Google Gemini for revenue predictions, demand forecasting, business health scores, and automated executive summaries.
- **Automated Reporting:** Reduce manual analytical overhead with system-generated reports for sales, finance, and inventory modules.
- **Seamless Integrations:** Connect natively with Zoho CRM, Books, and Inventory to ensure a single source of truth for all business data.
- **Data-Driven Decision Making:** Utilize predictive analytics to identify market trends, operational risks, and growth opportunities proactively.
- **Role-Based Security:** Ensure secure data access and compliance with customizable permissions tailored to specific organizational roles.

03. CUSTOMER PERSONA

Primary Persona: Operations Manager

Goal: Monitor and manage important business operations from one centralized platform while using real-time data and AI insights to make better decisions.

Needs & Goals

- Monitor sales performance and track inventory stock levels.
- Review financial performance and monitor employee activity.
- Identify operational risks and receive AI-powered recommendations.

Pain Points

- Business data spread across multiple systems leading to manual analysis.
- Time-consuming decision-making due to delayed reporting.
- Difficulty forecasting business performance and monitoring departments.

04. ARCHITECTURE & TECHNOLOGY

The ERP utilizes a modern frontend and backend architecture designed for scalability with direct AI and Zoho service integrations.

Technology Stack

Layer	Technology
Frontend	React.js, TypeScript, Tailwind CSS, Framer Motion
Backend	FastAPI, Python, Uvicorn, Pydantic
Database	PostgreSQL with SQLAlchemy ORM
Data Services	Zoho Books, CRM, and Inventory APIs
AI Integration	Google Gemini API
Infrastructure	Docker, Vercel, Railway

Architecture Flow

The system maintains a request-response lifecycle that bridges business data with AI analysis:

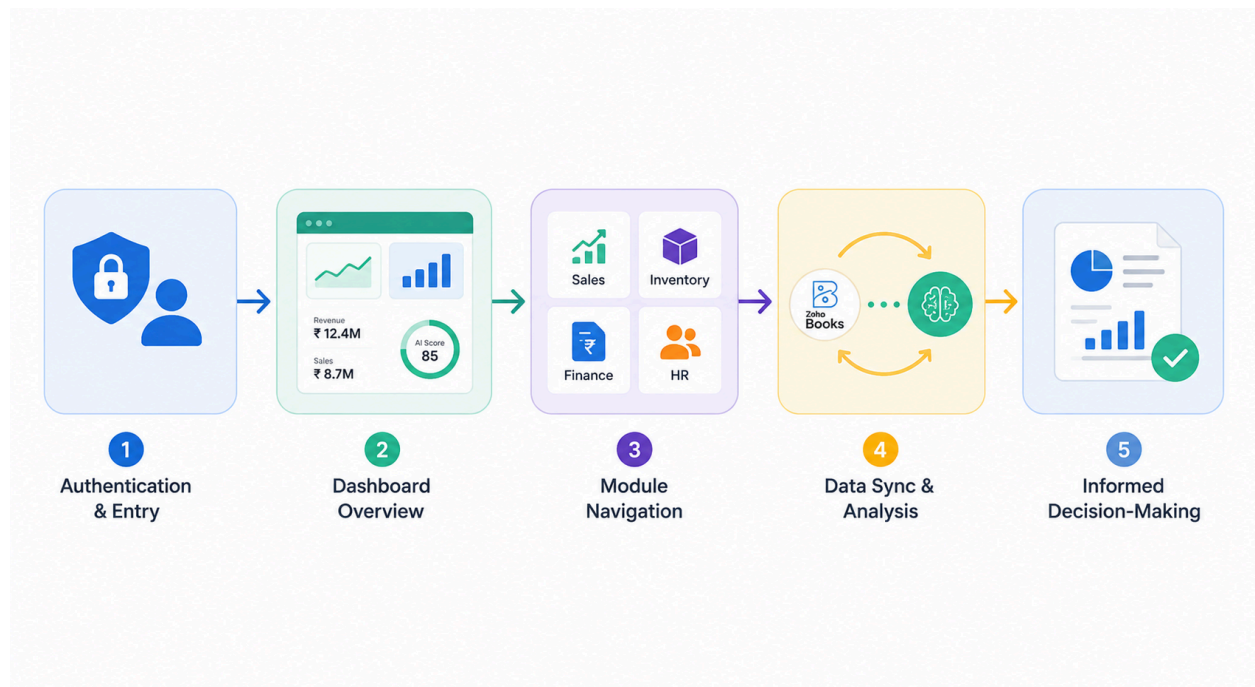
User → React → FastAPI → Zoho Books → Gemini AI → Insights

05. CORE ERP WORKFLOW

The workflow is designed around centralized business data, ensuring that external Zoho records are transformed into intelligent insights.

Operational Lifecycle

1. **Authentication & Entry:** Secure user sign-in and session management via JWT.
2. **Dashboard Overview:** Immediate visibility into KPIs including revenue, sales, and AI business score.
3. **Module Navigation:** Accessing Sales, Inventory, Finance, and HR management tools.
4. **Data Sync & Analysis:** Synchronizing Zoho Books data for processing by the AI Insights layer.
5. **Informed Decision-Making:** Utilizing executive summaries and growth forecasts to drive business strategy.



06. FEATURE MAP

Feature	Description
Dashboard	High-level overview of revenue, inventory value, and AI-generated health scores.
Sales Management	Processing orders, quotations, invoices, and sales performance forecasting.
Inventory	Stock level monitoring, warehouse tracking, and automated low-stock alerts.
Finance	Expense tracking, profit analysis, accounts payable, and accounts receivable.
Employee / HR	Directory management, department tracking, attendance, and leave payroll.
Reports	Exportable business reports (PDF/CSV) for sales, inventory, and finance.
AI Insights	Gemini-powered revenue prediction, churn analysis, and executive summaries.

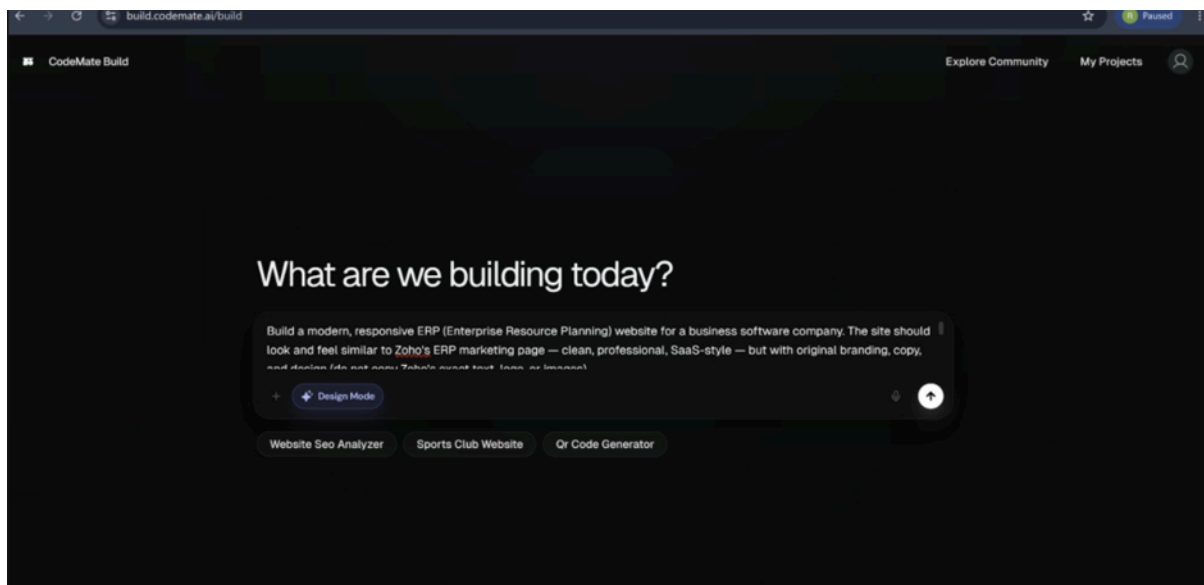
Feature	Description
Zoho Integration	Secure OAuth connection to synchronize external CRM and Books records.

07. BUILD: DESIGN MODE

Build Design Mode established a **premium enterprise SaaS interface** with a focus on dark aesthetics, glassmorphism, and interactive analytics.

Design Foundation

- **Prompt 1: Initial ERP Design** - Build a modern, responsive AI-powered ERP website for a business software company. Create a clean, premium SaaS experience with a professional marketing website and a complete ERP dashboard. Use purple and blue accents, modern charts, and a professional enterprise visual language.
- **Prompt 2: Module Cards & Analytics** - Create product-focused sections for the ERP that showcase sales, inventory, finance, and AI insights through embedded UI. Use interactive charts, glassmorphism components, and a strong visual hierarchy to demonstrate the platform's ability to centralize operational data.

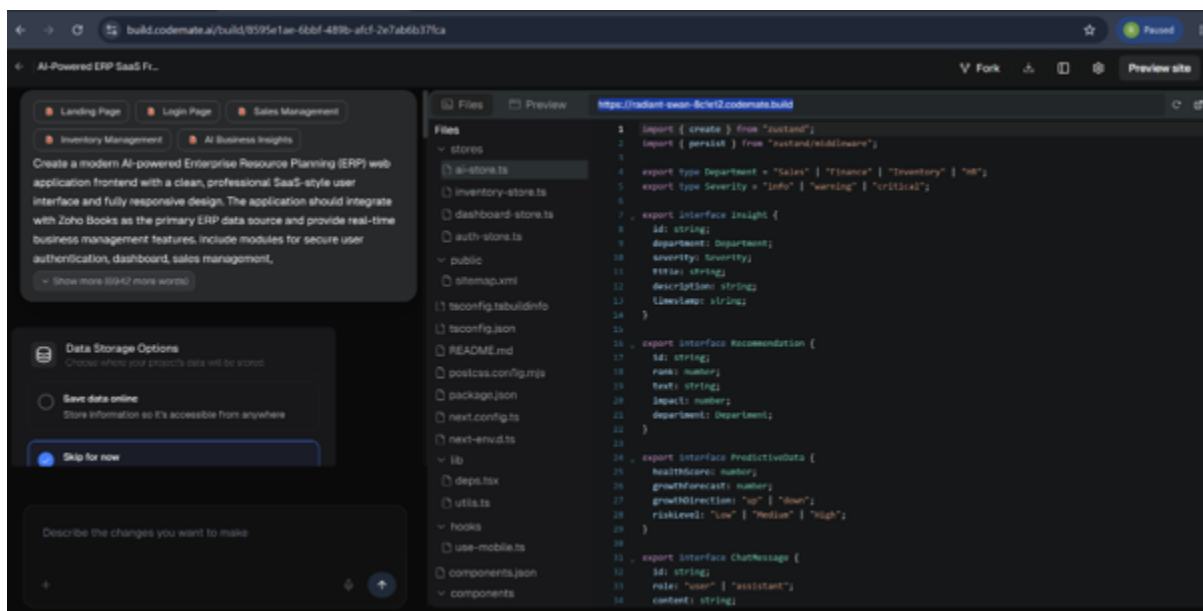


08. BUILD: CODE MODE

Transitioned approved designs into a functional frontend application with modular React components and mock business data.

Implementation Strategy

- **Prompt 1: Frontend Component Architecture** - Convert the approved ERP design into a responsive React application. Implement reusable components for sidebar, navbar, KPI cards, data tables, and AI panels. Preserve the dark premium interface and glassmorphism styling across all business modules.
- **Prompt 2: Business Module Prototypes** - Build functional modules for Dashboard, Sales, Inventory, and Finance. Include charts for revenue trends, order status tables, and inventory filters. Ensure the application is responsive across desktop, tablet, and mobile devices with proper navigation states.



09. BUILD PROMPT HIGHLIGHTS

The Build workflow focused on creating a production-ready ERP system with role-based access and Zoho integrations.

Representative Prompts

1. **ERP Modules & Dashboard:** Create a complete enterprise ERP dashboard with a collapsible sidebar and modules for Sales, Inventory, Finance, and AI Insights. Include KPI cards for revenue, orders, and inventory value, along with revenue trends and recent activities.
2. **AI & Zoho Integration:** Extend the ERP application with a dedicated AI Business Insights module powered by Google Gemini and a Zoho Integration Center. Include AI-powered health scores, revenue predictions, and OAuth-based Zoho connection management.
3. **Complete Functional ERP:** Implement reusable components, authentication, and role-based access across all modules. Ensure the application maintains a clean modular architecture and responsive layouts for desktop, tablet, and mobile devices.

10. CORA: PLANNING & IMPLEMENTATION

CORA served as the primary implementation layer, converting the React frontend into a backend-connected FastAPI application.

Implementation Workflow

Frontend → CORA Planning → FastAPI Backend → Zoho API → Gemini AI

Strategic Prompts

- **Prompt 1: Backend Architecture Setup:** Generate a production-ready FastAPI backend with modular API modules for authentication, dashboard, sales, inventory, and business insights. Use Python 3.11+, Pydantic, and JWT Authentication.
- **Prompt 2: Integration Services:** Implement the Zoho integration layer as a dedicated service. Connect Zoho Books as the primary source of business data and provide OAuth-based authentication and secure token handling.
- **Prompt 3: AI Insights Implementation:** Create a Gemini service that can analyze ERP data to generate health scores, revenue predictions, and automated executive summaries based on synchronized business records.

```
1 # Settings
2 from pydantic_settings import BaseSettings
3 from pydantic_settings import SettingsConfigDict
4
5 class Settings(BaseSettings):
6     app_name: str = "ERP Sphere"
7     app_version: str = "0.1.0"
8     environment: str = "development"
9     jwt_secret: str = "82aaf2067d14eb7e00d7e0ba81c17edca294c9c24f8a2a773d01effde71"
10     app_login_username: str = "admin"
11     app_login_email: str = "admin@ersphere.local"
12     app_login_password: str = "ERP2026!"
13
14     # Zoho Books uses OAuth 2.0. Create a self-client or server-based client in
15     # the Zoho API console and grant the listed scopes to obtain a refresh token.
16     zoho_books_base_url: str = "https://www.zohoapis.com/books/v2"
17     zoho_accounts_url: str = "https://accounts.zoho.com/oauth/v2/token"
18     zoho_client_id: str = "1000.F87DVGW6K139AUJ7NQ68MDLW094J8"
19     zoho_client_secret: str = "5aw4518b74f6f61912231c15048c2184545282"
20     zoho_refresh_token: str = "http://localhost:8080/oauth/callback?1000.F87DVGW6K139AUJ7NQ68MDLW094J8"
21     zoho_organization_id: str = "60878624842"
22     zoho_timeout_seconds: float = 30.0
23     zoho_use_mock_data: bool = True
24
25     # Application named persistence only. Zoho Books remains the source of
26     # truth for accounting and operational data.
27     database_url: str = "sqlite:///erp.db"
28     # A public OAuth client identifier. The matching JavaScript origins must also
29     # be configured in Google Cloud Console for Google Sign-In to work.
30     google_client_id: str = "712282122779-af4bdfaybhc4205ppm7hc7sig3d844.apps.googleusercontent.com"
31
32     model_config = SettingsConfigDict(
33         env_file=".env",
34     )
```

11. CORA TASK ARCHIVE

Category	Implementation Focus
Backend	FastAPI structure, modular API routing, and Pydantic validation.
Integrations	Zoho Books, CRM, Inventory OAuth, and manual/auto sync logic.
AI Services	Gemini-powered forecasting, risk analysis, and business health scores.
Security	JWT authentication, role-based access control, and protected routes.
Business Modules	Full CRUD for Sales, Inventory, Finance, Customers, and Suppliers.
Reports & Data	Interactive analytics charts, PDF/Excel exporters, and global search.

12. IMPLEMENTATION MILESTONES

Phase	Milestone	Focus Area
1	Foundation	Initial ERP experience design and React frontend development.
2	Architecture	FastAPI backend setup with modular service and API layers.
3	Auth & RBAC	JWT authentication and role-based access management.
4	Zoho Sync	Integration with Zoho Books, CRM, and Inventory data sources.
5	AI Insights	Gemini-powered business health analysis and forecasting.
6	Analytics	Reporting dashboards, data visualization, and final verification.

13. BACKEND, DATA & INTEGRATIONS

The ERP backend leverages a modular FastAPI architecture designed to maintain high cohesion across business domains including sales, inventory, and finance.

Core Integrations

- **Zoho Books:** Serves as the primary source of truth for business data including invoices, payments, and revenue tracking via OAuth-based synchronization.
- **Zoho Inventory:** Manages product stock levels, warehouse movement, and purchase orders associated with supplier interactions.
- **Google Gemini:** Powers the AI intelligence layer, analyzing synchronized business records to generate growth forecasts, expense analysis, and risk indicators.

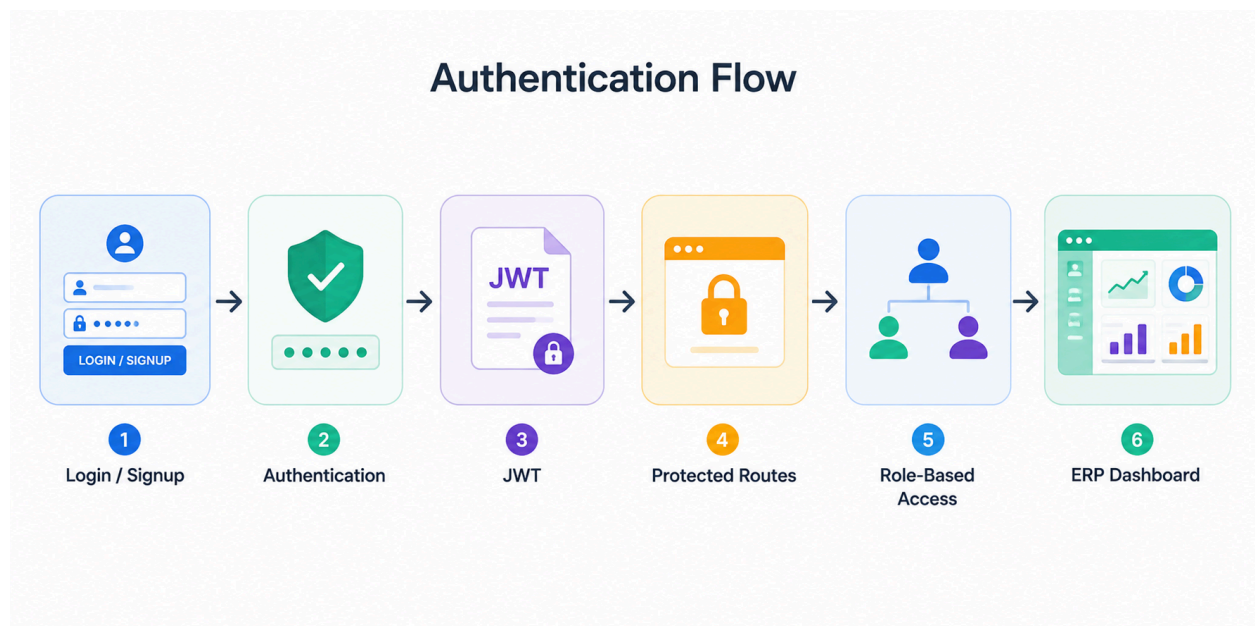
Data Pipeline

Business data flows from external Zoho services into the ERP backend service layer. The Gemini service then consumes this data to provide contextual analysis, which is visualized on the frontend dashboard as actionable business insights.

14. AUTHENTICATION & USER MANAGEMENT

The ERP platform includes a robust authentication layer with role-based access control (RBAC) to ensure secure data access for different departments.

Authentication Workflow



User Roles & Permissions

- **Admin:** Full access to settings, user management, and organization-level integrations.
- **Manager:** Access to dashboards, reports, analytics, and AI-generated recommendations.
- **Finance / Sales / Inventory:** Restricted access to their respective business modules and operational tools.

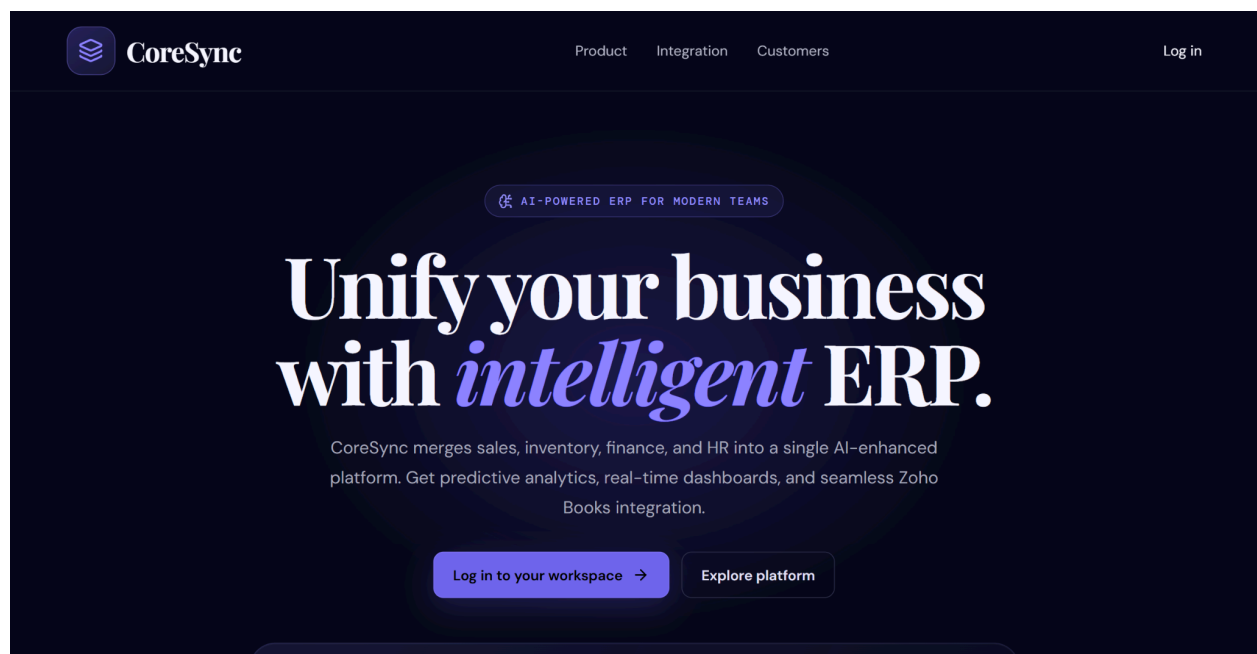
15. RESPONSIVE UI & POLISH

The ERP interface is built with a **Responsive First** principle, maintaining a premium enterprise SaaS experience across all device classes.

Detailed visual representations for each device layout are provided below.

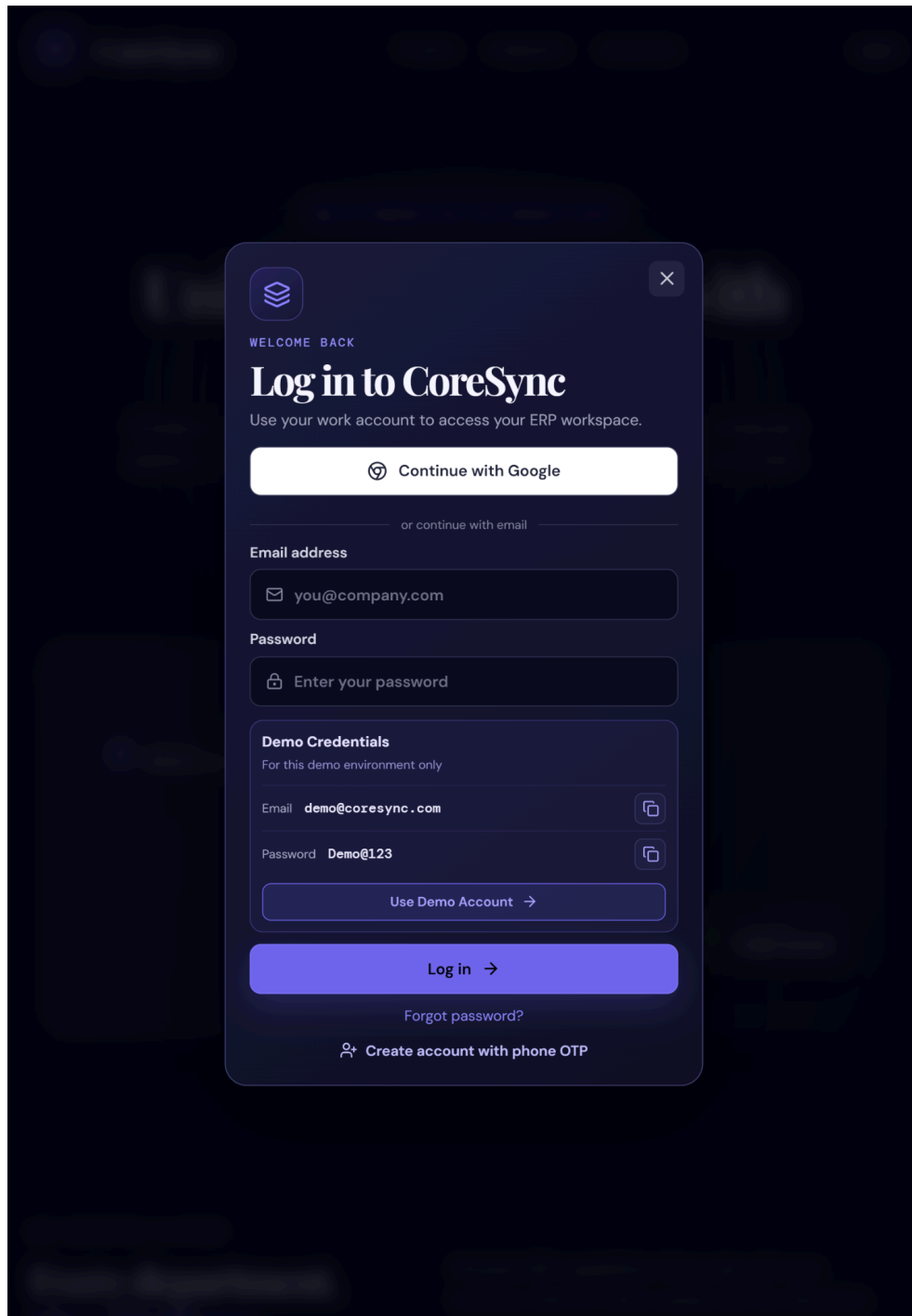
15.1 Desktop View

Features full sidebar navigation, multi-column business modules, and interactive analytics dashboards.



15.2 Tablet View

Utilizes adjusted navigation and optimized content density for responsive dashboard layouts.



15.3 Mobile View

Navigation collapses into a mobile-friendly menu with stacked layouts and touch-friendly interaction cards.

